

Chemical Equilibrium

Chapter

1

1.

2.2.9 (2015:04)

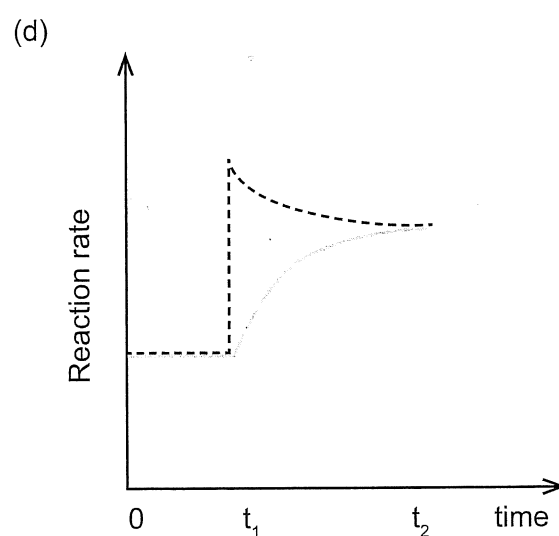
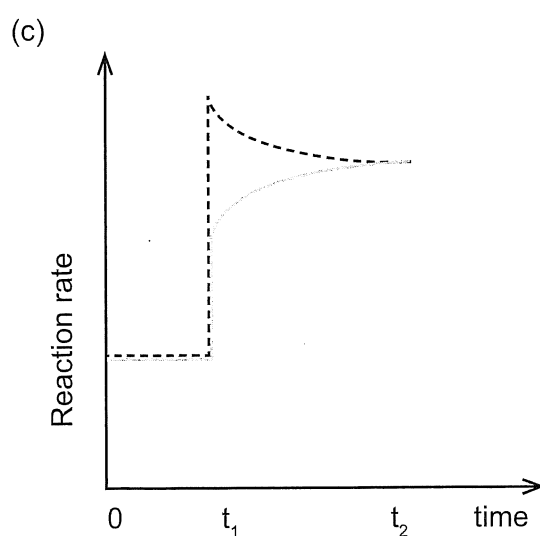
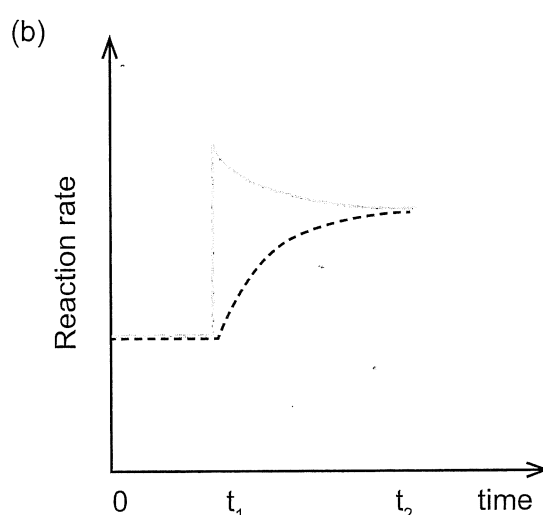
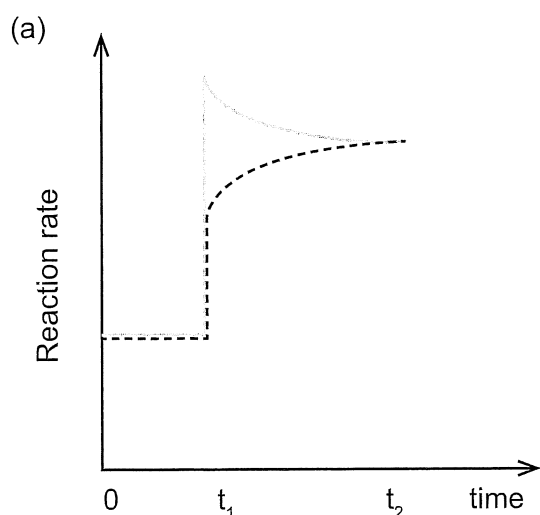
Consider the following equilibrium.



The system is initially at equilibrium. At time t_1 , the temperature of the system was increased. Which of the following **best** represents the changes in the forward and reverse reaction rates until equilibrium is re-established at time, t_2 ?

The forward reaction rate is represented by _____

The reverse reaction rate is represented by _____



Questions (2015:06) and (2015:07) refer to the reaction represented by the equation shown below.



2. 2.2.10 (2015:06)

Which one of the following is the equilibrium law expression for this reaction?

(a) $K = \frac{1}{[\text{H}^+]^4[\text{SO}_4^{2-}]^2}$

(b) $K = \frac{[\text{H}_2\text{O}]^2}{[\text{H}^+]^4[\text{SO}_4^{2-}]^2}$

(c) $K = \frac{[\text{PbSO}_4]^2}{[\text{H}^+]^4[\text{SO}_4^{2-}]^2}$

(d) $K = \frac{1}{[\text{H}^+]^2[\text{SO}_4^{2-}]}$

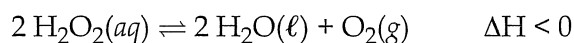
3. 2.2.9 (2015:07)

Assuming equilibrium has been established, which one of the following will cause a decrease in pH?

- (a) adding more solid lead
- (b) adding solid sodium sulfate
- (c) removing solid lead sulfate
- (d) adding barium nitrate solution

4. 2.2.9 (2016:01)

The decomposition of hydrogen peroxide in a closed system is represented by the equation below.



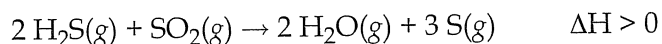
Which one of the following will increase the equilibrium yield of oxygen?

- (a) decreasing the concentration of hydrogen peroxide
- (b) increasing the total pressure of the system
- (c) decreasing the temperature of the system
- (d) adding an inert gas to the system

5.

2.2.1 (2016:02)

Sulfur can be obtained from hydrogen sulfide found in natural gas according to the equation below.



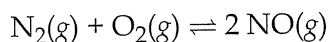
Which one of the following changes will initially decrease the rate at which sulfur is produced?

- (a) reduce the partial pressure of the hydrogen sulfide ($\text{H}_2\text{S}(\text{g})$)
- (b) increase the partial pressure of sulfur dioxide ($\text{SO}_2(\text{g})$)
- (c) add a metal catalyst to the reaction vessel
- (d) heating the reaction vessel

6.

2.2.11 (2016:03)

Consider the equilibrium system below.



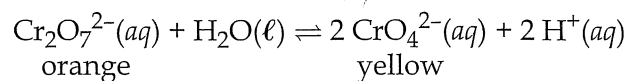
If the equilibrium constant (K) for this reaction is 4.1×10^{-31} , which one of the following statements is **true** for the system where the initial partial pressures of nitrogen and oxygen were equal to each other?

- (a) Once equilibrium is reached, the reverse reaction rate is much faster than the forward reaction rate.
- (b) The partial pressure of $\text{NO}(\text{g})$ is less than the partial pressure of $\text{N}_2(\text{g})$.
- (c) The actual ratio of gaseous N_2 particles to NO gaseous particles is 1:2.
- (d) When nitrogen gas is injected into a vessel containing mostly oxygen gas, the partial pressure of oxygen decreases dramatically.

7.

2.2.9 (2016:04)

A 500 mL solution of dichromate ions and chromate ions at equilibrium is described by the equation below.



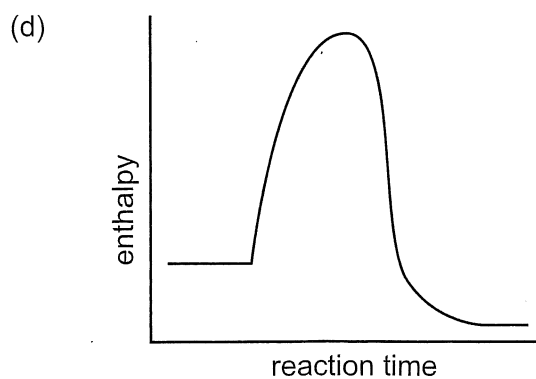
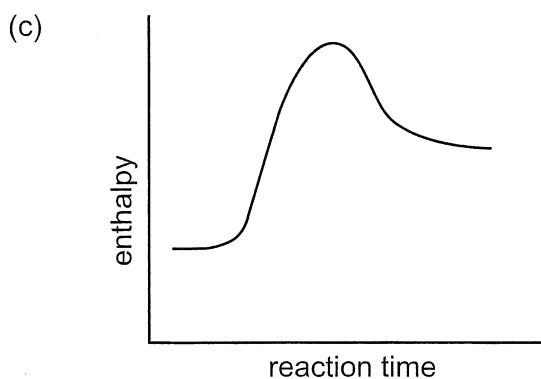
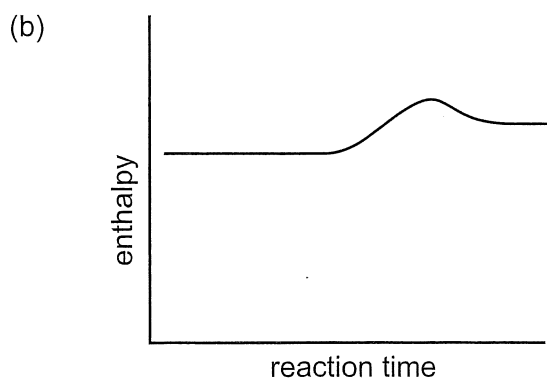
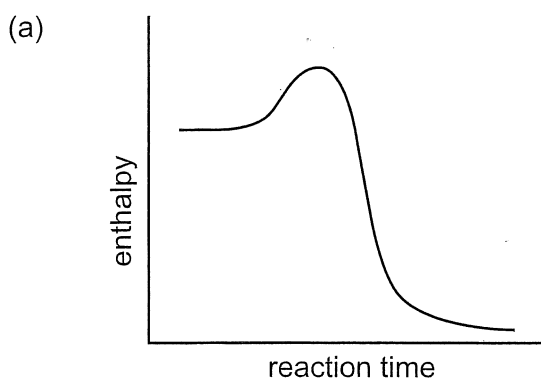
Which of the following **best** describes the effect of adding 10 mL of concentrated potassium hydroxide solution to the system once equilibrium has been re-established?

Relative change in concentration of $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$	Relative change in concentration of $\text{CrO}_4^{2-}(\text{aq})$	Relative change in concentration of $\text{H}^+(\text{aq})$	Colour change of solution
(a) increase	decrease	decrease	more orange
(b) decrease	increase	decrease	more yellow
(c) no change	no change	no change	no change
(d) decrease	increase	increase	more yellow

8.

2.2.7 (2016:08)

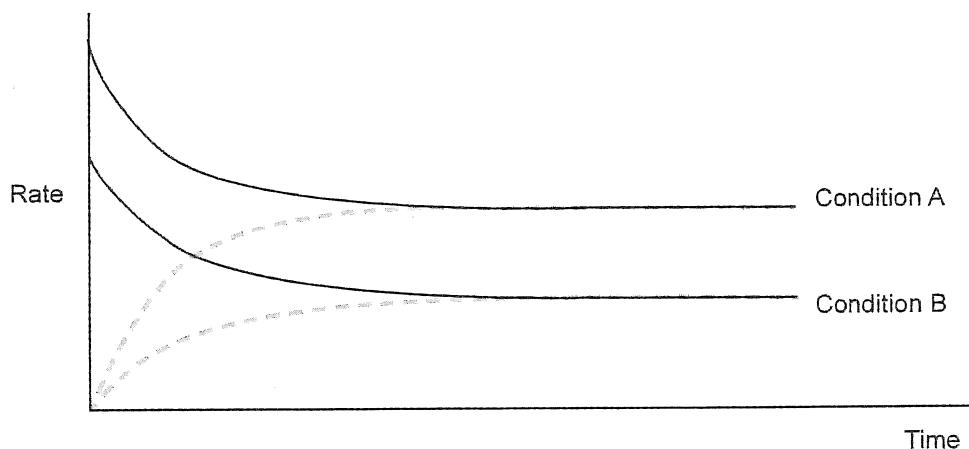
Which of the following energy profile diagrams **best** represents a spontaneous, exothermic reaction?



9.

2.2.1 (2017:05)

The following graph shows both the forward (—) and reverse (-----) reaction rates for a reaction under two different sets of conditions (Condition A and Condition B).



What is different about these conditions that would account for the graphs as shown?

- (a) Condition A is at a higher temperature.
- (b) Condition B is at a higher temperature.
- (c) Condition B had a catalyst added to it.
- (d) Condition B was conducted at a higher temperature and pressure.

10.

2.2.1 (2017:07)

Consider the following statements about the effect of a catalyst being added to a reaction mixture.

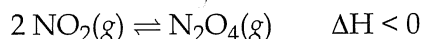
- (i) The formation of intermediate species not found in the uncatalysed reaction.
- (ii) The availability of a new reaction path having a lower activation energy.
- (iii) An increase in the percentage of collisions resulting in a reaction.
- (iv) The equal increase of both forward and reverse reaction rates.

Which of these statements describe what will occur when a catalyst is added to a reaction mixture?

- (a) i and ii only
- (b) ii and iv only
- (c) i, ii and iv only
- (d) i, ii, iii and iv

11. 2.2.9 (2017:12)

A sealed glass tube at room temperature contains nitrogen dioxide (a brown gas) and dinitrogen tetroxide (a colourless gas) in equilibrium, as represented by the following equation.



If the appearance of the gas mixture at room temperature is pale brown, which one of the following is true if the glass tube is placed in hot water?

- (a) The gas mixture will not undergo any noticeable change in appearance.
- (b) The gas mixture will become darker brown.
- (c) The gas mixture will become even paler at first, but would then return to its original appearance of pale brown.
- (d) The gas mixture will become colourless.

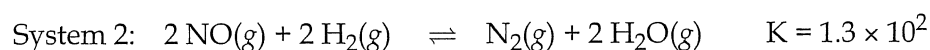
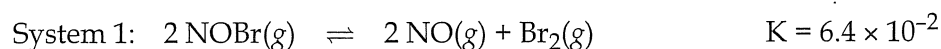
12. 2.1 (2017:18)

The acidification of oceans due to their increased concentrations of carbon dioxide decreases the rate and amount of calcification in some marine organisms, e.g. shellfish and coral reefs. Which one of the following equations best represents the chemistry involved in decreasing the rate and amount of calcification?

- (a) $2 \text{H}^+ + \text{CaCO}_3 \rightarrow \text{Ca}^{2+} + \text{H}_2\text{O} + \text{CO}_2$
- (b) $\text{CO}_2 + \text{H}_2\text{O} + \text{CO}_3^{2-} \rightarrow 2 \text{HCO}_3^-$
- (c) $4 \text{H}^+ + 2 \text{CO}_3^{2-} \rightarrow \text{H}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
- (d) $\text{CO}_2 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

13. 2.2.11 (2018:02)

Consider the following reaction systems at equilibrium.

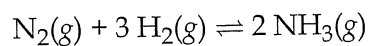


Which of these statements regarding these systems is/are true?

- (i) System 2 reaches equilibrium faster than System 1.
 - (ii) The greatest ratio of products to reactants occurs in System 2.
 - (iii) Equilibrium in System 1 favours the reactants more than it does in System 2.
- (a) i only
 - (b) ii only
 - (c) i and iii only
 - (d) ii and iii only

14.

The Haber Process involves the following equilibrium reaction:



A number of closed reaction vessels were set up containing the gases shown in the table below.

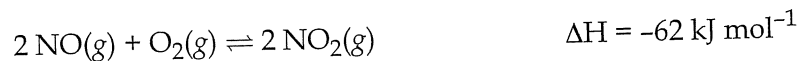
Reaction vessel	Gases initially present
i	nitrogen, hydrogen
ii	nitrogen
iii	ammonia
iv	hydrogen, ammonia

In which of the above closed reaction vessels would equilibrium be established after a period of time?

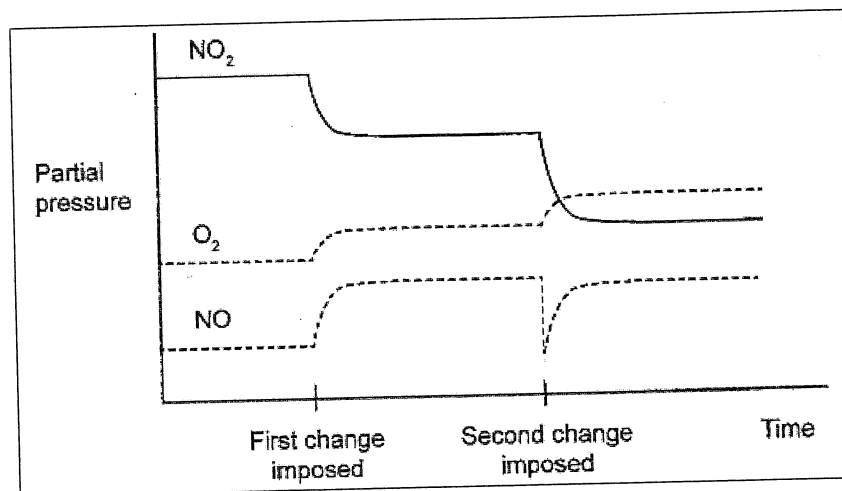
- (a) i only
- (b) i and iii only
- (c) i, iii and iv only
- (d) ii, iii and iv only

Questions (2019:03) and (2019:04) refer to the following information.

Nitrogen dioxide, $\text{NO}_2(\text{g})$, is formed when nitrogen monoxide, $\text{NO}(\text{g})$, undergoes oxidation as shown below.



A change was imposed on an equilibrium gas mixture of NO_2 , NO and O_2 . The mixture returned to equilibrium and another change was imposed. The following graph shows the effects of the two changes.



15.

2.2.9 (2019:03)

Identify the imposed changes that best account for the shape of the graph.

	First change	Second change
(a)	the temperature is decreased	the partial pressure of O_2 is increased
(b)	the temperature is decreased	the partial pressure of NO is decreased
(c)	the temperature is increased	the partial pressure of O_2 is increased
(d)	the temperature is increased	the partial pressure of NO is decreased

16.

2.2.11 (2019:04)

What do the initial partial pressures of the three gases indicate?

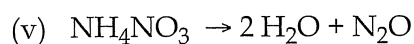
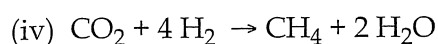
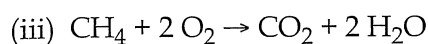
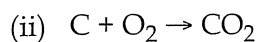
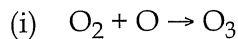
- (a) The relative proportions of the gases present at equilibrium.
- (b) That there is initially no NO gas present in the system.
- (c) That the NO_2 gas reaches equilibrium first.
- (d) That the O_2 and NO gases are producing NO_2 at a faster rate than they are being formed.

17.

2.1 (2019:12)

The United Nations Kyoto Protocol and the Intergovernmental Panel on Climate Change aim to secure a global commitment to reducing greenhouse gas emissions over the next few decades.

Which of the following equations shows the production of a greenhouse gas?



(a) i and ii only

(b) ii and iii only

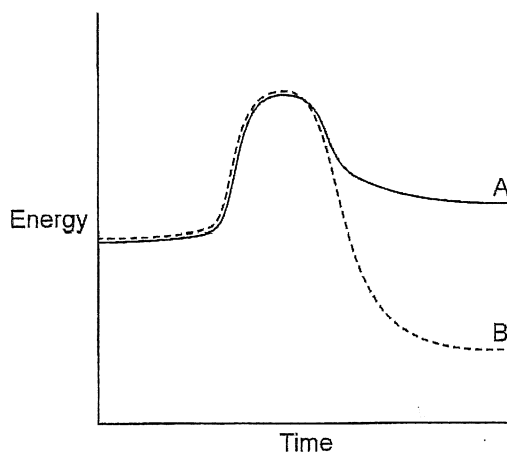
(c) iii, iv and v only

(d) i, ii, iii, iv and v

18.

2.2.6 (2020:08)

Energy profile diagrams for two different chemical reactions (A and B) are shown below.



Which of these reactions is the more likely to be reversible and why?

(a) Reaction A, because its forward reaction is endothermic.

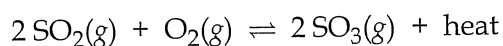
(b) Reaction B, because its forward reaction is exothermic.

(c) Reaction A, because the activation energy of its reverse reaction is smaller than that for Reaction B.

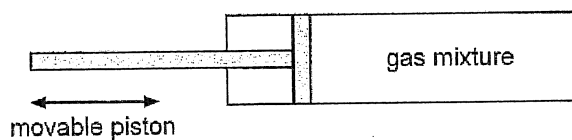
(d) Neither, because the activation energies of their forward reactions are the same.

21.

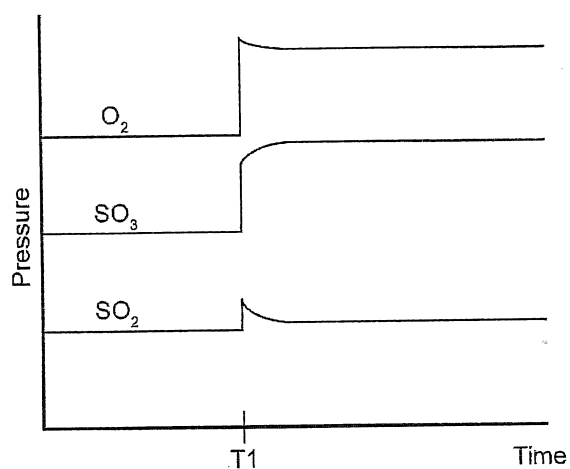
Consider the following equilibrium:



A mixture of these gases was at equilibrium in a sealed container with a movable piston, as shown below.



A change was applied to the system at T1 and the results of this change are shown in the following graph.



What was the change that occurred at T1?

- (a) An inert gas was added to the reaction vessel.
- (b) The reaction vessel was heated.
- (c) A catalyst was added to the reaction vessel.
- (d) The volume of the reaction vessel was decreased.

Chapter 1: Chemical Equilibrium

1. (2015:04) Answer 'c'. Remember the graph is showing Rate – both increase with temperature and then a new equilibrium is established.
2. (2015:06) Answer 'a'. $K = \frac{[\text{products}]}{[\text{reactants}]}$. We do not include species with (s) or (l).
3. (2015:07) Solid lead does not appear in the K expression. Adding Na_2SO_4 will increase pH. Lead sulfate is a solid. Answer 'd'. Barium sulfate is insoluble and $\text{Ba}^{2+}(\text{aq})$ removes sulfate ions moving the equation to the LHS.
4. (2016:01-c) The reaction is exothermic – cooling the system would give more oxygen as the reaction moves to the RHS. Adding inert gas would have no effect, as the partial pressure of oxygen does not change. Decreasing the conc of H_2O_2 would lead to a lower forward rate and less O_2 . Increasing the pressure would increase the partial pressure of oxygen which would result in less O_2 forming. The answer is 'c'.
5. (2016:02-a) Note the question asks for decrease in rate. Answers 'b', 'c' and 'd' always increase reaction rate. Decreasing the partial pressure of H_2S will decrease the frequency of collisions and thus the rate of formation of sulfur. The answer is 'a'.
6. (2016:03-b) Answer 'a' is wrong. At equilibrium the forward and reverse reaction rates are equal. Answer c would suggest a much larger K value. Answer d would be correct for a large Kc reaction. The value of Kc is very small. Thus $[\text{products}]/[\text{reactants}]$ is small i.e. there are few products. The answer is 'b'.
7. (2016:04-b) This is a classic question. If you are told something not in the equation is added, be sure it will probably react with something in the equation either as acid/base or a precipitate. The added solution is concentrated so there is no need to worry about diluting the reactants and products. Here the H^+ is destroyed so the equation moves to the right – the answer is 'b'.
8. (2016:08-a) The words spontaneous, exothermic means a reaction with a small E_a and a negative ΔH . The answer is 'a'.
9. (2017:05-a) The question is based on the fact that all reactions have a higher rate at higher temperatures. The answer is 'a'.
10. (2017:07-d) This question covers the multiple properties of catalysts that you should know. All are valid. The answer is 'd'.
11. (2017:12-b) Another LCP question. The addition of heat will favour the endothermic reaction which is the reverse reaction. The answer is 'b'.
12. (2017:18-b) This question is taken directly from the syllabus 'Acidification shifts the equilibrium of carbonate chemistry in seawater, decreasing the rate and amount of calcification'. Answer (b) shows the destruction of carbonate ions that are essential to the process of calcification. The answer is 'b'.
13. (2018:02-d) The magnitude of a K value indicates the product:reactant ratio. It says nothing about the rate of reaction. Therefore answer (i) is incorrect. Both (ii) and (iii) are true. The answer is 'd'.
14. (2019:02-c) In two places the question mentions that the system is closed. Therefore, equilibrium can be established if either
 1. all the reactants are present,
 2. all the products are present
 Therefore, item ii cannot establish equilibrium. The answer is 'c'.
15. (2019:03-d) At the first change the partial pressures slowly moved to new equilibrium positions indicating a new K value. Thus, it must have been a temperature change. As the reaction is exothermic and the partial pressure of NO_2 decreased the reaction must have been heated – answer 'c' or 'd'.
At the second point the partial pressure of NO decreased dramatically and then partly recovered, shown by (b) or (d). The answer is 'd'.
16. (2019:04-a) This question investigates the concept of equilibrium and could have been used to show that $K > 1$ for this reaction as at equilibrium the equation lies to the right-hand side. The question, however, as asked the answer is 'a'.
17. (2019:12-d) A greenhouse gas is any gas with more than two atoms. Some such as SF_6 are 23 000 times more effective than CO_2 but they are all greenhouse gases. Thus, these are all greenhouse gases – the answer is 'd'.
18. (2020:08) The magnitude of the reverse activation energy determines whether a reaction is reversible. Endothermic reactions have smaller reverse E_a than the forward E_a . The answer is (c).
19. (2020:12) A closed system allows energy to enter or leave but not matter. This is a closed system and by applying heat the equilibrium position was changing – this is a chemical process but it results in a colour change – a physical process. The answer is (c).
20. (2020:16) An equilibrium reaction does not go to completion so answer (a) is possible but unlikely. Answers (b) and (c) are correct as the reaction has not stopped and the forward and backward rates are equal. Kc is temperature dependent so (d) is correct. The answer is (a).
21. (2020:24) At T1 the partial pressure of each of the gases increased proportionally. This shows that the total volume decreased increasing the partial pressures. The reaction then moved to the right to overcome this constraint and re-establish equilibrium. The answer cannot be (a) as the partial pressures would not change, nor (b) as the reaction would move to favour one side, nor (c) as there would no change at T1 as the system was already at equilibrium. The answer is (d).

Chapter 2: Acids and Bases

1. (2015:15-d) Answer 'd'. Irrespective of the strength of an acid or base the mol ratio does not change.
2. (2015:16-b) Answer 'b'. $\text{pH} < 0$ suggests $[\text{H}^+]$ of $> 1 \text{ mol L}^{-1}$ $\text{pH} + \text{pOH} = 14$.
3. (2015:18-d) Answer 'd'. HCO_3^- gains a proton in the forward reaction and HS^- gains one in the reverse reaction.
4. (2015:19-c) Answer 'c'. All demonstrate change in pH with hydrolysis except NaCl – a neutral salt.